

AFRILANG Stdlib

std/c/mlgradient

Bibliothèque AFRILANG `std/c/mlgradient` (niveau complex, catégorie complex). Elle expose 7 fonction(s) :
gradSum, gradM

```
`import "std/c/mlgradient" . `use mlgradient`
```

- `gradSum(v list number) ? number`
- `gradMean(v list number) ? number`
- `gradMin(v list number) ? number`
- `gradMax(v list number) ? number`
- `gradVar(v list number) ? number`
- `gradStd(v list number) ? number`
- `gradNorm(v list number) ? list number`

Category: complex | Tier: complex | Functions: 7

FRANCAIS

1. Vue d'ensemble

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```
`import "std/c/mlgradient"` . `use mlgradient`  
- `gradSum(v list number) ? number`  
- `gradMean(v list number) ? number`  
- `gradMin(v list number) ? number`  
- `gradMax(v list number) ? number`  
- `gradVar(v list number) ? number`  
- `gradStd(v list number) ? number`  
- `gradNorm(v list number) ? list number`
```

2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/mlgradient"  
use mlgradient
```

Niveau : complex · Catégorie : Modules complex (c/)
Domaines avancés ? graphes, NLP, réseaux complexes.

3. Référence API

```
? gradSum(v list number) ? number  
? gradMean(v list number) ? number  
? gradMin(v list number) ? number  
? gradMax(v list number) ? number  
? gradVar(v list number) ? number  
? gradStd(v list number) ? number  
? gradNorm(v list number) ? list
```

4. Exemples d'utilisation

```
import "std/c/mlgradient"  
use mlgradient  
  
create result = gradSum(/* args */)   
say result  
  
create result = gradMean(/* args */)   
say result  
  
create result = gradMin(/* args */)   
say result  
  
create result = gradMax(/* args */)   
say result  
  
create result = gradVar(/* args */)   
say result  
  
create result = gradStd(/* args */)   
say result
```

5. Bonnes pratiques

- ? Préférez ``import "std/c/mlgradient"`` en tête de fichier.
- ? Lancez ``afirilang check`` avant de livrer.
- ? Combinez avec les modules `stdlib` de la même catégorie.
- ? Voir `/stdlib/` pour le source live et les démos playground.
- ? Signalez les problèmes sur GitHub si une export manque.

6. Annexe ? source

```
module mlgradient
```

```
function gradSum(v list number) returns number
end

function gradMean(v list number) returns number
end

function gradMin(v list number) returns number
end

function gradMax(v list number) returns number
end

function gradVar(v list number) returns number
end

function gradStd(v list number) returns number
end

function gradNorm(v list number) returns list number
end
end
```

ENGLISH

1. Overview

AFRILANG library `std/c/mlgradient` (tier complex, category complex). Exports 7 function(s):
gradSum, gradMean, gradMin,

```
`import "std/c/mlgradient"` . `use mlgradient`  
- `gradSum(v list number) ? number`  
- `gradMean(v list number) ? number`  
- `gradMin(v list number) ? number`  
- `gradMax(v list number) ? number`  
- `gradVar(v list number) ? number`  
- `gradStd(v list number) ? number`  
- `gradNorm(v list number) ? list number`
```

2. Installation & import

Add the module to your program:

```
import "std/c/mlgradient"  
use mlgradient
```

Tier: complex · Category: Complex modules (c/)
Advanced domains ? graphs, NLP, complex networks.

3. API reference

```
? gradSum(v list number) ? number  
? gradMean(v list number) ? number  
? gradMin(v list number) ? number  
? gradMax(v list number) ? number  
? gradVar(v list number) ? number  
? gradStd(v list number) ? number  
? gradNorm(v list number) ? list
```

4. Usage examples

```
import "std/c/mlgradient"  
use mlgradient  
  
create result = gradSum(/* args */)   
say result  
  
create result = gradMean(/* args */)   
say result  
  
create result = gradMin(/* args */)   
say result  
  
create result = gradMax(/* args */)   
say result  
  
create result = gradVar(/* args */)   
say result  
  
create result = gradStd(/* args */)   
say result
```

5. Best practices

```
? Prefer `import "std/c/mlgradient"` at the top of your file.  
? Run `afirang check` before shipping.  
? Combine with related stdlib modules from the same category.  
? See /stdlib/ for the live source and playground demos.  
? Report issues on GitHub if an export is missing or undocumented.
```

6. Appendix ? source

```
module mlgradient
```

```
function gradSum(v list number) returns number
end

function gradMean(v list number) returns number
end

function gradMin(v list number) returns number
end

function gradMax(v list number) returns number
end

function gradVar(v list number) returns number
end

function gradStd(v list number) returns number
end

function gradNorm(v list number) returns list number
end
end
```